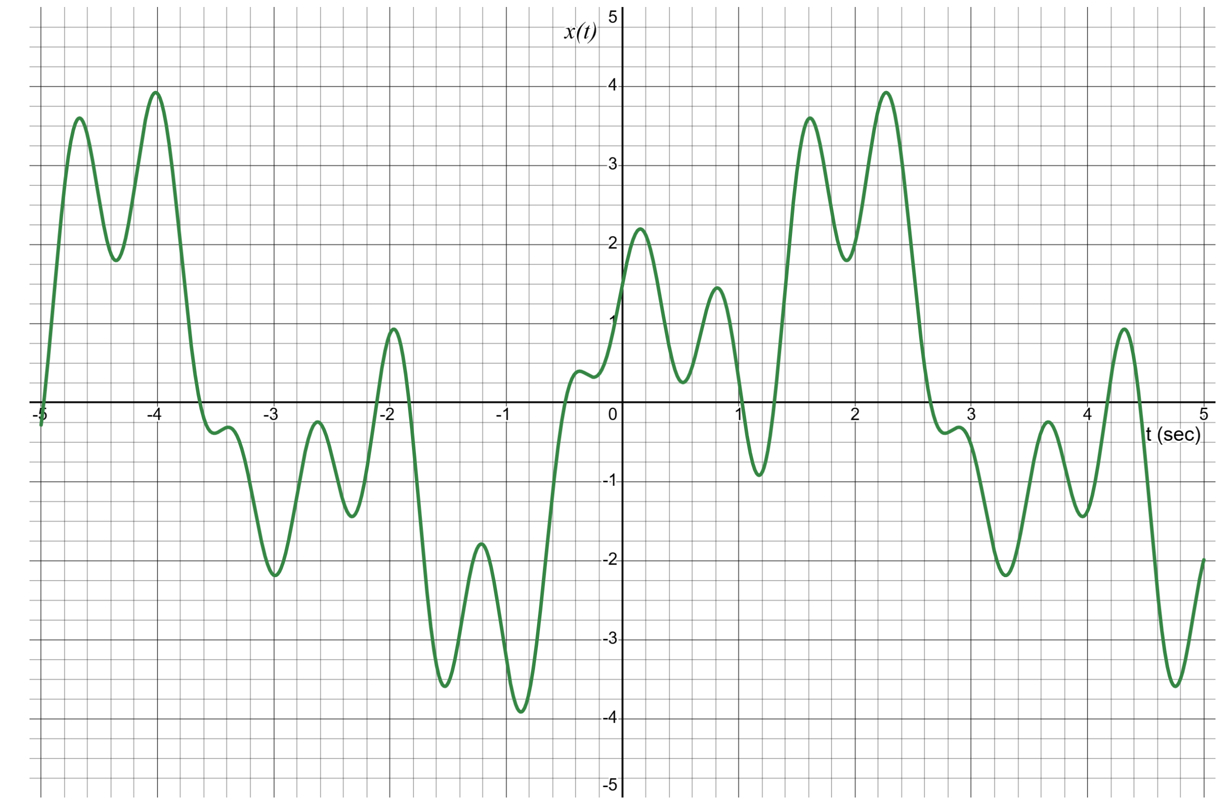
Brac University
Semester: Fall 2025
Course Code: CSE430
Digital Signal Processing
Section: 1
Faculty: EHQ

Assessment: *Assignment 1*
Release Date: October 21, 2025
Submission Deadline: November 4, 2025
Full Marks: 45

- ✓ The assignment must be submitted within the deadline. Late submissions will be penalized.
- ✓ Submission of softcopy of the assignment is mandatory.
- ✓ Submit your assignment through this link - https://forms.gle/B35eTsZvePytm9YK8

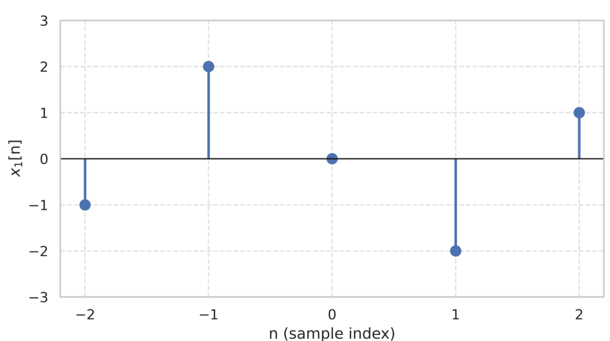
■ Question 1 of 3 *[10 marks]*

Sample the signal at a frequency of 5 Hz. Print out the signal for your ease. You can sample on the same plot.



■ Question 2 of 3 *[20 marks]*

Draw the outputs of the provided systems. Also figure out the outputs as functions of the inputs.



$x_1[n]$

(1)

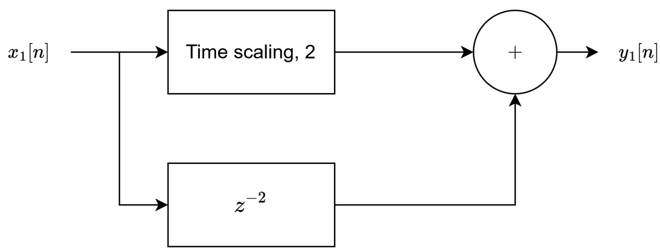
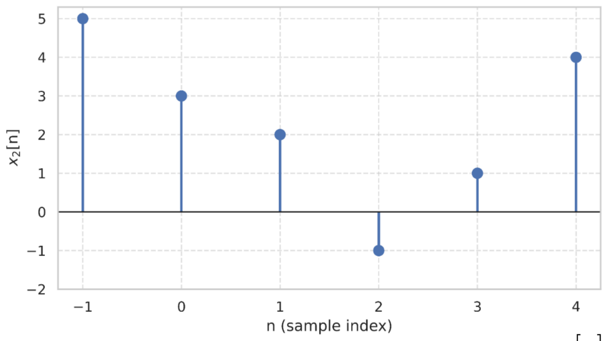


Figure 1: System 1



$x_2[n]$

(2)

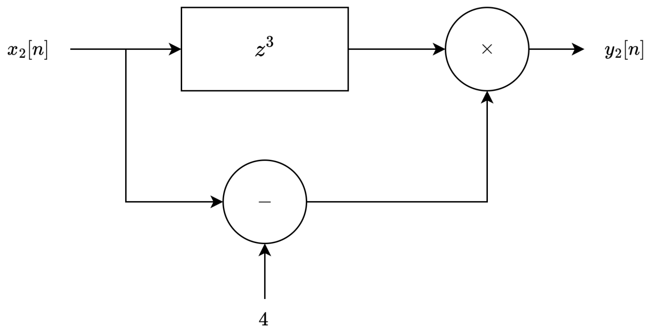
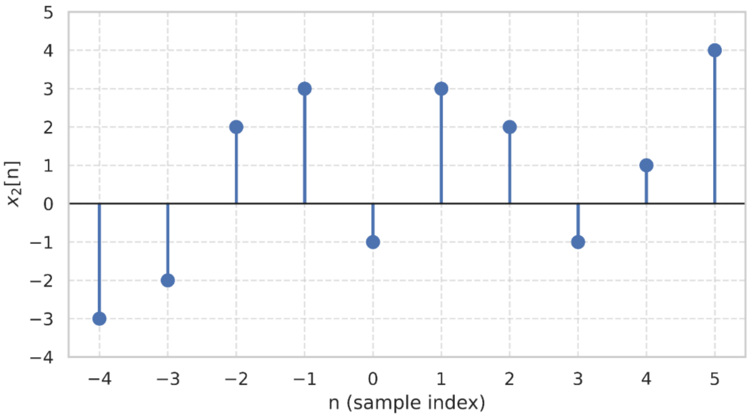


Figure 2: System 2

$x_3[n]$



(3)

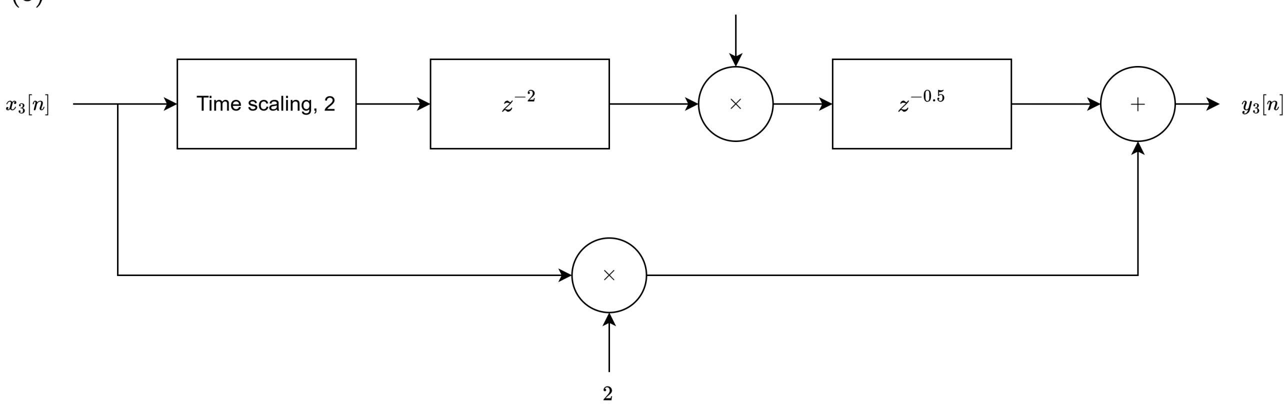


Figure 3: System 3

■ Question 3 of 3 *[15 marks]*

Three different systems are provided in the question. Considering how the systems update the input signals, design the systems using simple blocks - time scaling, time shifting, amplitude scaling and amplitude shifting.